

PrimeSim™ HSPICE® What's New

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What's New in This Release

The Synopsys PrimeSim HSPICE Release Notes provide information on new features, enhancements, changes, known issues, and resolved Synopsys Technical Action Requests (STARs). To view the release notes, go to [My Product Releases](#) on SolvnetPlus, locate and click **PrimeSim HSPICE**, and then click the release version of interest.

New Features, Enhancements, and Changes

This release includes the following new features, enhancements, and changes.

- [Analysis Enhancements](#)
 - [Device Model Enhancements](#)
 - [DP/Cloud Enhancements](#)
 - [MOSRA Enhancements](#)
 - [Performance Enhancements](#)
 - [Signal Integrity Enhancements](#)
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Analysis Enhancements

Added the following analysis enhancements:

- Integrated the initialization and main sensitization into a single simulation in PrimeSim HSPICE.
- Added a new option `.option enforce_sensitivity_analysis` to enforce the sensitivity table generation in post process of Monte Carlo analysis (that is, AVA report), if independent random variables (IRV) number is more than 50,000. For more information, see `.OPTION ENFORCE_SENSITIVITY_ANALYSIS` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.

- Added the support of expressions and `get_measval()` function in `.DATA` command in PrimeSim HSPICE. For more information, see `.DATA` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Implemented the high/low glitch detection feature based on the vector file in PrimeSim HSPICE.

Device Model Enhancements

Added the following enhancements for device models:

- Added the SiPH related features: optical probing, MTS/3DIC, and so on.
- Enabled thermal modeling in general subckt-based macro model for power devices. For more information, see Simulation of Thermal Effect in Macro Models in the *PrimeSim™ HSPICE® User Guide: Basic Simulation and Analysis*; and `.SET_NODE_TYPE`, `.OPTION ENABLE_SUBCKT_SHE` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Added the following new compact models and revisions:
 - BSIM-SOI 100.1. For more information, see LEVEL 70: BSIMSOI Model in the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.
 - HiCUM L2 3.1.0. For more information, see LEVEL=8 HICUM Model in the *PrimeSim™ Continuum Reference Manual: Device Models*.
 - HiSIM-SOTB 1.3. For more information, see LEVEL 84: HiSIM-SOTB in the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.
 - MVSG 4.0. For more information, see LEVEL 82: MVSG Model in the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.
 - ASM-HEMT 104
 - BSIM-CMG106.1+TMEMOD2.1
 - BSIM-CMG106.1+TMEMOD2.2
 - BSIM-CMG106.1+TMEMOD GIDL equation update
 - BSIM-IMG102.96+Aging
 - HiSim-HMG update (Beta 5)
 - HiSim-HV 2.5.1
 - PSP104
 - UTSOI 102.8

- Added support for the RRAM compact models in PrimeSim HSPICE.
- Enhanced `MOSRAPRINT` to support `temp` as input.

DP/Cloud Enhancements

Optimized the CDSL application in PrimeSim HSPICE DP for better performance in large Monte Carlo simulations.

MOSRA Enhancements

Added the following MOSRA enhancements:

- Added support to run the MOSRA-based LTE simulation and aging simulation in one SPICE deck in PrimeSim HSPICE.
- Added support for the `RDRMUE` aging simulation in HiSIM-HV 2.1. For more information, see MOSRA API Supported Aging Degradations in the *PrimeSim™ MOSRA User Guide*.
- Enhanced the examples to illustrate the usage of user-defined degradation types. For more information, see Gradual Aging Analysis and MOSRA API Supported Aging Degradations in the *PrimeSim™ MOSRA User Guide*.

Performance Enhancements

Added support for the consistent syntax of bisection with PrimeSim:

- Implemented the same window syntax as in PrimeSim.
- Implemented the one-side search algorithm, to improve the bisection performance.

Signal Integrity Enhancements

Added the following Signal Integrity enhancements:

- Enhanced the edge superposition to capture the jitter amplification accurately in StatEye. For more information, see `.STATEYE` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.
- Added the IFFT causality enforcement feature into SPUTIL to enhance the `Causality` function. For more information, see `.CAUSALITY` in the *PrimeSim™ Continuum Reference Manual: Commands and Control Options*.

Chapter 1: What's New in This Release

New Features, Enhancements, and Changes

- Added an internal switch to use PrimeSim HSPICE transient engine for StatEye.
- Updated the PrimeSim HSPICE Touchstone demo files, which have significant warnings/errors.